

ARTIFICIAL INTELLIGENCE MINI PROJECT

SmartCare AI – Hospital Triage & Emergency Routing

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Class	TYIT
Division	A
College Name	Sheth L.U.J. College of Arts and Sir M.V. College of Science and Commerce
Academic Year	2026-2027
Subject	Artificial Intelligence
Faculty Name	Chetna Tanvade

Project documentation prepared according to the supplied AI Mini Project Documentation Format.

1. Cover Page

• **Project Title • Student Name • Roll Number • Class • College Name • Academic Year • Subject**

SmartCare AI – Hospital Triage & Emergency Routing

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2. Introduction

Briefly explain: • What is the project? • What problem does it solve? • Why is this project useful?

SmartCare AI is an Artificial Intelligence mini project for hospital triage and emergency ambulance routing. It combines intelligent-agent concepts, search algorithms, Bayes' rule, machine-learning classifiers and a fairness check. The system helps triage patients according to symptoms and demonstrates how an ambulance route can be selected from a city-road graph. It is useful because it demonstrates how AI can support urgent decision-making while keeping clinicians in control.

3. Problem Statement

Clearly mention the problem that the project aims to solve.

In emergency healthcare, patients need to be assigned an appropriate urgency level and ambulances need efficient routes to hospitals. The project addresses this problem by modelling a table-driven triage agent, comparing BFS, DFS, UCS, Greedy and A* search for ambulance routing, applying Bayes' rule to diagnostic-test interpretation, and evaluating machine-learning models for urgency classification.

4. Objectives

Write 3 to 5 objectives of the project.

To develop a SmartCare AI prototype for hospital triage and emergency routing.

To apply BFS, DFS, UCS, Greedy and A* search algorithms to an ambulance-routing graph.

To demonstrate Bayes' rule and probability-based reasoning in a healthcare setting.

To train and compare classification models for patient urgency prediction.

To examine fairness and the effect of class-weight balancing on recall.

5. Technologies Used

Mention the technologies/software used.

HTML, CSS and JavaScript for the interactive SmartCare AI interface.

SVG-based visualization for the city-road/ambulance route map.

Machine-learning classification models: Logistic Regression, Naive Bayes, Decision Tree and kNN.

5-fold cross-validation and feature scaling for model evaluation.

Bayesian-network work with pgmpy, as listed in the project interface.

GitHub/project documentation as a reference source.

6. Dataset

Explain: • Dataset name • Source of dataset • Number of records • Important attributes/features • Target variable

Dataset name: SmartCare AI synthetic patient dataset.

Source: The supplied project states that the patient data is synthetic; no external dataset source is specified in the supplied project.

Number of records: 2,000 patient records.

Important features: The supplied project describes patient records being used for urgency classification and symptom-based triage; a complete column-by-column feature list is not provided in the supplied HTML.

Target variable: Patient urgency level/classification used by the urgency models.

7. Methodology / Working

Explain the steps followed in the project:

1. Create a hospital emergency-room triage scenario and a city-road graph for ambulance routing.
2. Define a table-driven triage agent using symptoms such as chest pain, low oxygen, high fever and cough.
3. Apply BFS, DFS, UCS, Greedy and A* to search from Andheri to Hospital.
4. Demonstrate Bayes' rule using disease rate, test sensitivity and false-alarm rate.
5. Prepare 2,000 patient records for classification using a 75/25 train-test split; missing values are filled with the median and features are scaled.
6. Train and compare Logistic Regression, Naive Bayes, Decision Tree and kNN.
7. Evaluate accuracy, recall, F1 and ROC-AUC, with 5-fold cross-validation for Logistic Regression.
8. Perform a fairness check by group and examine class-weight balancing.

8. Algorithm Used

Mention the AI/ML algorithm used. Briefly explain: • What the algorithm is • Why it is used in the project • How it works

BFS: Breadth-First Search explores paths level by level. It is useful for comparing an uninformed route-search strategy.

DFS: Depth-First Search explores one branch deeply before backtracking. It demonstrates a different uninformed search behaviour.

UCS: Uniform Cost Search expands the lowest accumulated-cost path and is suitable for finding a minimum-cost route when edge costs represent distance.

Greedy Search: Uses a heuristic estimate toward the hospital to choose the next promising path.

A*: Combines path cost and heuristic estimate ($g + h$), providing a cost-aware heuristic search.

Logistic Regression: A classification model used for predicting urgency classes. In this project it achieved the best reported test accuracy and ROC-AUC among the four models.

Naive Bayes: A probabilistic classifier based on Bayes' rule and conditional-independence assumptions.

Decision Tree: A tree-based classifier that makes decisions through feature-based splits.

kNN: Classifies a record using nearby training examples; the project uses $k=7$.

Reported routing comparison from the supplied SmartCare AI route graph:

Algorithm	Route	Distance (km)	Cities explored	Stops
BFS	Andheri → Bandra → Dadar → Sion → Hospital	13.1	11	4
DFS	Andheri → Bandra → Dadar → Kurla → Powai → Thane → Chembur → Sion → Worli → Colaba → Hospital	35.9	11	10
UCS	Andheri → Bandra → Dadar → Sion → Hospital	13.1	11	4
Greedy	Andheri → Bandra → Dadar → Sion → Hospital	13.1	5	4
A*	Andheri → Bandra → Dadar → Sion → Hospital	13.1	6	4

9. Implementation

Include: • Important Python code • Explanation of the code • Screenshots of code/output if required

The supplied project is implemented as an interactive HTML application. JavaScript creates the road graph, performs the search algorithms, updates the route visualization, implements the table-driven triage agent, and calculates the Bayes posterior probability.

Important implementation logic for routing: the project stores city coordinates and weighted road edges, maintains a frontier of candidate paths, applies the selected search priority, avoids already explored nodes, and returns the path when Hospital is reached.

Important implementation logic for triage: chest pain produces a critical ambulance/cardiologist action; low oxygen produces a critical oxygen/doctor action; high fever with cough produces an urgent isolation/testing action; high fever alone is moderate; cough alone is mild; no listed symptom is routine.

Bayes implementation calculates posterior probability as $\text{sensitivity} \times \text{disease probability}$ divided by the sum of true-positive and false-positive probabilities.

The interface also reports model metrics and a fairness comparison.

Representative JavaScript snippet:

```
const r = search("A*");
const c = cost(r.p);
sCost.textContent = c.toFixed(1);
pathTxt.textContent = r.p.join(" → ");
```

10. Results

Show the final results of the project. Include: • Accuracy/performance • Prediction results • Graphs • Output screenshots

The best-performing reported classifier is Logistic Regression with 81.4% test accuracy, recall 0.54, F1 score 0.61 and ROC-AUC 0.85.

Naive Bayes achieved 76.2% accuracy, recall 0.44, F1 0.50 and ROC-AUC 0.81.

Decision Tree achieved 78.8% accuracy, recall 0.44, F1 0.53 and ROC-AUC 0.77.

kNN ($k=7$) achieved 75.6% accuracy, recall 0.41, F1 0.48 and ROC-AUC 0.74.

Logistic Regression 5-fold cross-validation averaged 82.0% accuracy.

Class-weight balancing increased overall recall from 0.54 to 0.75.

Fairness check: the supplied project reports recall of 0.50 for male patients and 0.64 for female patients, a gap of 0.14, above the stated 0.10 review threshold.

The route comparison shows A* selecting Andheri → Bandra → Dadar → Sion → Hospital with a reported distance of 13.1 km and 6 cities explored.

Model performance table:

Model	Test accuracy	Recall	F1	ROC-AUC
Logistic Regression	81.4%	0.54	0.61	0.85
Naive Bayes	76.2%	0.44	0.50	0.81
Decision Tree	78.8%	0.44	0.53	0.77
kNN (k=7)	75.6%	0.41	0.48	0.74

11. Advantages

Write 3 to 5 advantages of the proposed system.

Combines multiple AI techniques in one healthcare-oriented application.

Compares several route-search algorithms instead of relying on one method.

Provides an interpretable table-driven triage response for common emergency symptoms.

Allows direct comparison of multiple urgency-classification models using several performance metrics.

Includes a fairness check and demonstrates mitigation using class weights.

12. Limitations

Write 2 to 4 limitations of the project.

The patient dataset is synthetic, so performance may not represent real hospital populations.

The supplied project does not provide a complete feature/column description for the 2,000 records.

The model is a support tool and should not replace clinical judgement or emergency medical protocols.

The reported fairness gap requires further investigation and broader validation across patient groups.

13. Applications

Mention where the system can be used in real life.

Emergency-room patient triage support.

Ambulance route planning and comparison of search strategies.

Decision-support systems for prioritizing patients by urgency.

Educational demonstrations of AI search, probability, agents and machine learning.

Fairness auditing of healthcare AI models before deployment.

14. Conclusion

Write a short summary of: • What was developed • What was achieved • How AI was used in the project

SmartCare AI was developed as a healthcare-focused AI mini project combining intelligent agents, search, Bayesian reasoning, machine learning and fairness analysis. The project demonstrates emergency triage rules, compares five route-search strategies, and evaluates four urgency-classification models. Logistic Regression produced the strongest reported classification performance, while class-weight balancing improved recall. Overall, the project shows how AI can support emergency decision-making while emphasizing that clinical oversight, privacy and fairness checks remain important.

15. Future Scope

Mention how the system can be improved in the future.

Validate the models using larger, real-world and clinically reviewed datasets while maintaining patient privacy.

Add richer patient features and a documented data dictionary.

Use live traffic and hospital-capacity information for dynamic ambulance routing.

Improve fairness monitoring with more demographic and clinical groups and regular audits.

Integrate the prototype with secure hospital systems and clinician feedback workflows.

16. References

Mention books, websites, datasets, articles, or documentation used.

SmartCare AI supplied project source (interactive HTML project).

GitHub/project documentation referenced in the supplied project.

pgmpy, for Bayesian-network related project work as identified in the supplied interface.

No external patient dataset was specified; the supplied project states that patient data is synthetic.

End of AI Mini Project Report